

"I Feel It Like I Actually Feel It": How HCI Can Support the Experiences of Synaesthesia

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Abstract

Synaesthesia is a cross-sensory phenomenon where the stimulation of one sense involuntarily triggers another sensory percept. Previous research in HCI often engaged with synaesthesia in conceptual or metaphorical terms, overlooking the experiences of synaesthetes themselves. We sought to understand the experiences of synaesthesia as a means to inform cross-sensory interaction design. We interviewed 12 synaesthetes and analysed their experiences using Interpretative Phenomenological Analysis (IPA). We found that synaesthetes often struggled to articulate their experiences due to limited sensory vocabulary, time requirements, and a lack of understanding from others. Our findings contribute to an understanding of synaesthesia that embraces its diversity in all forms and moves beyond the immediate perceptual experiences of synaesthesia and toward a more holistic reflection of such experiences as they are embedded in self-reflection, community and environments; and how the insights derived from this preliminary research can guide interaction designers in creating more inclusive technology to support individuals with synaesthesia.

Keywords

Synaesthesia, Cross-sensory Interaction, Interpretative Phenomenological Analysis

ACM Reference Format:

Mamoru Watanabe, Oussama Metatla, and Atau Tanaka. 2025. "I Feel It Like I Actually Feel It": How HCI Can Support the Experiences of Synaesthesia. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3720202>

1 Introduction

'I was talking to a friend of mine who is a music producer about what kind of music I want to do. And then I started describing everything in shapes and colours. He was like, do you have synaesthesia? And I was like, what is that?'

This quote illustrates how one of our interview participants, referred to as P3, first learned that her unique perceptual experiences could be identified as synaesthesia. Following this discovery, P3 decided to take an online standardised test to validate her experiences.

However, the test primarily focused on musical pitch, which did not align with her form of Auditory-Visual synaesthesia. As a result, she was not classified as a synaesthete. This outcome discouraged her from pursuing further exploration of synaesthesia, leading to a seven-year gap before she revisited the topic.

This narrative highlights a critical disconnect between medicalised frameworks of synaesthesia and the nuanced experiences of individuals who identify as synaesthetes.

Synaesthesia has been widely studied across diverse fields, spanning scientific disciplines such as psychology and neuroscience, as well as creative domains including visual art, music, and literature [11, 16, 35, 36]. It is estimated that approximately 4.4% of the population experiences some form of synaesthesia [8]. The types of synaesthesia reported vary significantly, with estimates suggesting between 75 and 300 distinct forms, depending on the criteria used for assessment [26, 30, 34]. However, although a range of scientifically objective evaluation tests exists [3, 6, 9], these methods often prioritise advancing psychology or neuroscience research rather than supporting individuals with synaesthesia. These quantitative approaches typically reflect a medicalised and fail to address the personal experiences and needs of synaesthetes.

We propose that qualitative methods in Human-Computer Interaction (HCI) present an opportunity to engage deeply with the experiences of synaesthetes, fostering a more inclusive and empathetic understanding of their needs. While interest in synaesthesia within HCI is growing, much of the existing research has treated the topic metaphorically, frequently lacking direct engagement with synaesthetes themselves. This oversight limits the ability of HCI to authentically incorporate their insights and address their unique perspectives in the design of interactive technology. Our work seeks to advance synaesthesia research within HCI by prioritising direct engagement with synaesthetes, placing their experiences at the centre of the research process. By bridging the gap between the top-down, medicalised conceptualisations of synaesthesia as a neurological condition and the bottom-up, diverse experiences of synaesthetes, we aim to cultivate a more comprehensive and inclusive understanding of synaesthetic phenomena and its place in HCI.

As a first step, we present preliminary findings from an interview study that aimed to address two research questions:

- Q1. How do synaesthetes experience synaesthesia?
- Q2. How can these synaesthetic experiences inform HCI?

We begin by providing background on synaesthesia, offering an overview of psychology research in the field and reviewing how synaesthesia has been addressed within the domain of HCI. Next, we present a qualitative interview methodology we used to explore and understand individual synaesthetic experiences. The outcomes

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720202>

of these interviews are analysed to identify key themes that emerge across participants. In the discussion, we address how our findings contribute to an understanding of synaesthesia that embraces its diversity in all forms and moves beyond the immediate perceptual experiences of synaesthesia and toward a more holistic reflection of such experiences as they are embedded in self-reflection, community and environments; and how the insights derived from this preliminary research can guide interaction designers in creating more inclusive systems to support individuals with synaesthesia. Our future work building on these preliminary findings centres around a series of co-design workshops currently being conducted with synaesthetes. These workshops, along with the insights gained from the interviews, aim to contribute to the development of synaesthetic interaction design that integrates and supports the unique experiences of synaesthetes.

2 Background

2.1 Synaesthesia

The word ‘Synaesthesia’ combines the Greek roots *syn* (together) and *aesthesia* (sensation) [33] and is often described as ‘mixing’ [32], ‘merging’ [12], or ‘union’ [35] of the senses. The range of descriptions and definitions creates difficulty in arriving at a single understanding of synaesthesia. Due to the ambiguity surrounding the definition of synaesthesia, there are no standardised measurements. More than 75 types of synaesthesia have been reported, with the most common being *grapheme to vision*, which includes *grapheme-colour synaesthesia*, where individuals perceive colours associated with letters [26]. Some individuals experience phantom tastes when hearing, speaking, or reading words, a form known as *lexical-guided synaesthesia* [1]. There are also those who visualise speech as internally translated letters, a form referred to as *Tickertape synaesthesia* [10].

There are two common ways in which synaesthesia manifests: *associator* and *projector* types. An associator-type experiences internal sensations (i.e., imagining the colour in their mind’s eye), while a projector-type experiences external sensations (i.e., seeing the colour projected on top of a letter).

Another approach to categorising synaesthesia is to differentiate between *developmental* and *acquired* synaesthesia. *Developmental* synaesthesia refers to forms that emerge in early childhood and are often linked to genetic or neurodevelopmental factors. Although synaesthesia is frequently viewed as a congenital condition in medical contexts, research suggests that it can also be *acquired* in adulthood [4]. Furthermore, there have also been distinctions between *strong* and *weak* synaesthesia; with strong synaesthesia referring to vivid sensory percepts triggered by another sensory stimuli; and weak synaesthesia reflecting cross-sensory correspondences expressed through language, perceptual similarity, or interactions during information processing [17]. Synaesthesia is often assessed for its ‘genuineness’ by measuring the consistency of synaesthetic experiences. However, Proulx and Stoerig have questioned the reliability of consistency as a criterion for qualifying a sensory experience as synaesthesia. They suggest that ‘the genuineness of synaesthesia does not necessarily need to be reflected in the stability of association’ [22]. As a way to address the ambiguity surrounding

the concept, Deroy and Spence advocate for a distinction between cross-modal correspondence and synaesthesia [5].

2.2 Synaesthesia in HCI

Multisensory and cross-sensory interactions are well-defined topics in HCI. Multisensory interaction considers the integration of a wide range of the human senses to design interactive systems [20], while cross-sensory interaction aims to ground such endeavour in the specific study of crossmodal cognition [18] and sensory substitution [15, 25, 31]. However, few research within these strand in sensory HCI research consider synaesthesia as an explicit research focus, and instead engage with synaesthesia in concetual or metaphorical terms. Lee et al. developed a card-type ideation toolkit for designers and HCI researchers inspired by synaesthesia. This toolkit aimed to enhance the idea of technology by illustrating connections between two sensory modalities [14]. The Synesthesia Suit was originally developed as an external device for the PlayStation VR game called ‘Rez Infinite’. This suit generates haptic vibrations based on the visual and audio cues from the game, providing an immersive experience [13]. Ward et al. created an electric nose that converts smells into real-time visual information in the form of 2D abstract shapes [24]. Wirfs-Brock et al. organised a CHI workshop engaging synaesthesia as a phenomenon to explore ways in which sensory translation might give ideas for data representation [37]. Neuwirth introduces the concept of ‘legal synaesthesia’ using synaesthesia as a metaphor to explore how AI technologies can create new legal experiences and emphasise the need for adaptable legal frameworks to address the evolving relationship between AI and law [19]. Wright and Kefalidou propose a ‘synaesthetic-oriented design’ inspired by the condition as a resource for user experience design in mixed reality environments [38]. O’Toole draws upon what he calls the ‘theory of synaesthesia’ to aid in understanding user associations across sound and image in multimodal interaction, noting “the possibility of the condition being present in everyone.” [21] In digital health research, SYNCalc, is a customisable calculator application that allows users to assign different colours to numbers, catering specifically to individuals with grapheme-colour synaesthesia [29]. While this work leverages grapheme-colour synesthesia in device interaction, it remains limited in its coverage, as it does not fully address the diverse spectrum of synaesthesia.

In these cases, synaesthesia serves as a reference point for heightened sensory experiences that span across perceptual modalities. It is utilised as a design resource to create interactions intended for general users. However, this approach frames synaesthesia as a “tool” rather than recognising it as an experiential phenomenon experienced by a subset of the population who identify as having the condition.

3 Study: Exploring synaesthetic experiences

As outlined above, synaesthesia research typically falls into two categories: a medicalised perspective, which acknowledges the existence of synaesthesia but offers limited practical applications, and metaphorical interpretations, which emphasise its heightened multisensory and cross-sensory nature but often fail to directly engage with synaesthetes and their experiences. There is therefore a significant gap in research that focuses on centring the experiences of

synaesthetes in the design of interactive technologies that support their needs. In the work presented here, we aim to prioritise the experiences of synaesthetes as a foundational step toward designing technologies that address their unique requirements. To achieve this, we conducted an interview study guided by two research questions: 1) What are the experiences of synaesthesia? and 2) How can synaesthetic experiences inform HCI research?

3.1 Participants

We recruited 12 participants (nine females and three males) aged 21 - 42 years old. The participants' backgrounds included eight North Americans, three Europeans, and one Asian. In total, the participants reported 30 different types of synaesthesia, with most experiencing multiple forms. The most common type was Grapheme-Colour (P1, P2, P4, P5, P6, P7, P9, P10, P12), followed by Spatial-Sequence (P1, P2, P4, P5, P10), Auditory-Visual (P1, P3, P5, P10), Mirror-Touch (P1, P11, P12) and Tickertape (P1, P10, P12). Some less common forms included Concept-Shape (P3), Sexual Synaesthesia (P12), and Sound-Personification (P12). Participants were recruited through social media, word of mouth, and dedicated mailing lists. The study was cleared by the authors' institutional ethics committee. We were conscious that we were not fully embedded members of the synaesthete community. While the first author identifies as potentially being a synaesthete, the other two authors do not identify as synaesthetes.

3.2 Procedure

We conducted the interviews both online and in person. Three participants attended in person, while nine participated online. For in-person interviews, we used a designated room at the authors' institution. Online interviews were conducted using Zoom Workplace and Microsoft Teams. We've recorded all interviews (audio and video), with each session averaging an hour in length. Before the interviews, we asked participants to complete a pre-interview questionnaire. This form collected detailed information, including their name, age, background, gender, types of synaesthesia, and responses to two specific questions: 1) 'How certain are you that you have synaesthesia?'; 2) 'Could you briefly share one of your most vivid (strong) synaesthetic experiences?' We gathered and reviewed these questionnaires prior to the first interview to guide us in tailoring the interview questions.

We conducted the interviews in a semi-structured format, preparing a range of questions to explore personal synaesthetic experiences. Before finalising the interview questions, we conducted a pilot test to refine our approach.

During the sessions, we asked participants similar sets of questions, which allowed them to elaborate on each topic and freely discuss their synaesthetic experiences. The interviews covered several key topics: participants' general understanding of synaesthesia, their past experiences (including how they discovered their synaesthesia), and both the positive and negative aspects of living with the condition. During the conversations, we encouraged participants to share and reflect on their experiences in detail. To elicit their synaesthetic experiences during the interview, we asked all participants 'If possible, could you demonstrate your synaesthesia

now?' Each participant brought and engaged with their own sensory materials such as scents, observing the letters on a display, or playing sounds, to describe their synaesthetic experience. Their description included verbal explanations, and some participants visualised their experiences by drawing what they perceived on paper using coloured pens.

3.3 Interpretive Phenomenological Analysis

Recognising the highly individual nature of our participants' stories and experiences, we refined our methodology to capture the unique, experiential aspects of what they sought to convey. To achieve this, we selected Interpretive Phenomenological Analysis (IPA). IPA is a qualitative research method that avoids the reductionist tendency of generalising findings. Instead, IPA emphasises how individuals make sense of their personal experiences [7]. Simmonds-Moore previously employed IPA in psychology research with synaesthetes [27], providing an effective model for studying such nuanced phenomena. In HCI, researchers have also used IPA to elicit individual narratives in socially sensitive settings. For example, Arvola and Linder applied IPA when working with migrants in Scandinavia, each with their own traumatic experiences [2]. Similarly, Snyder et al. employed IPA to explore the experiences of family caregivers in dementia research [28], while Rapp and Boldi used it to analyse users of persuasive technologies, assessing the personal impact of behavior change strategies [23]. These studies, like ours, involved small sample sizes and highly personal stories, yet their findings informed interaction design research. We followed Simmonds-Moore's psychological study, which used IPA to examine the "exceptional experiences" of a single synaesthete [27]. Their process of extracting themes from interview transcripts informed both our interview approach and subsequent analysis. However, we adapted their approach to 1) work with a group of synaesthetes; and 2) with HCI applications in view.

4 Findings

The process of IPA analysis requires engaging deeply with each individual case, followed by integrating all case analyses to identify connections across cases. We began by examining the responses of each participant to gain a detailed understanding of their personal experiences. After completing individual analyses, we analysed the responses of all 12 participants collectively, grouping responses that addressed similar themes or issues.

We used techniques from IPA to progress from specific personal themes to higher-level themes. This process followed three key steps: 1) Starting with Experiential Statements, 2) Identifying Personal Experiential Themes (PETs), and 3) Distilling Group Experiential Themes (GETs). In the following, the raw responses from participants served as the foundation for Experiential Statements and for extracting PETs, capturing their unique experiential insights. We then present the identified commonalities and recurring patterns which we synthesise as GETs. Using this approach we, derived three overarching Group Experiential Themes (GETs), which we outline below: *Importance of community and environment*; *Impact on personal life and instrumentation*; *Importance of self-reflection over time*. Our key finding is thus that centering these individual experiences reveals that synaesthetic experiences, while rooted

in heightened cross-sensory perceptions, extend far beyond these immediate sensory phenomena. Synaesthesia is shaped by and, in turn, influences broader aspects of life, including community, environmental factors, and ongoing self-reflection over time. Here, we present the broad groupings that emerged iteratively from our analysis to provide a holistic perspective on the synaesthetic experience. These groupings illustrate the complexity and richness of synaesthesia, as it intertwines with various dimensions of individual and social life.

4.1 Importance of community and environment

All participants described how synaesthesia can be an isolating experience and expressed a strong desire to connect with broader communities of synaesthetes to foster a sense of belonging. Given the rarity of this condition in the general population, this sense of isolation was particularly acute for some participants. As P3 shared: *"[it's] so cool to speak to someone who is studying synaesthesia, because it's not a very common topic and, to be honest, I'm very isolated here in Japan there is literally nobody that I know who knows what synaesthesia is."*

Participants emphasised the importance of sharing their experiences with others, noting the mutual benefits for both the individual synaesthete and the community. For example, P3 recounted how connecting with another synaesthete and learning their vocabulary helped her better articulate her own experiences: *"She also has a lot of different forms of synaesthesia, and I found her account and there are all these resources. And I was like, 'Ohh, there's all this vocabulary that she's giving me that I can use to describe my synaesthesia,' because before that I didn't talk about it because, well, I didn't know any synaesthetes around me."* However, participants also highlighted the challenges of sharing their experiences due to the lack of open and supportive environments. P6 reflected on the difficulty of initiating conversations about synaesthesia: *"I would have to raise the subject from nowhere. Talk about it exclusively from my perspective, and then no one might be able to relate to it or even understand what I'm trying to convey."*

These experiences suggest that certain environments and communities are more conducive to openness and sharing of experiences, as well as to the elicitation of synaesthetic experiences themselves. Some participants noted that their synaesthetic perceptions could be influenced by the structure of their physical and social environments. For example, P12 shared how minimising sensory distractions enhanced her ability to experience synaesthesia clearly: *"If I want to hear a sound or see it very clearly, if there isn't any sense, no touch, no itchy clothes, and if there's really just this one thing, then I see it clearer. And it doesn't really have to be open or closed eyes. But [if] it really gets chaotic, and I don't know what it is."* Similarly, P11 emphasised the importance of a comfortable environment in facilitating synaesthetic experiences: *"I think it's going to be very difficult because if you're in an environment where you're not comfortable, like I know for me, when I discovered it, I was in the confines of my own home, I was doing my own thing."* These accounts may suggest that synaesthetic experiences are highly influenced by both physical and social environments. The ability to engage with synaesthesia meaningfully depends not only on individual factors

but also on the external context and the presence of supportive, understanding communities.

4.2 Impact on personal life and instrumentation

Participants described the significant impact synaesthesia has on their lives, highlighting both the advantages and challenges of the condition. Many emphasized the importance of developing coping mechanisms and learning to harness their synaesthesia to improve their daily experiences and overall outlook on life. For instance, P11, who has mirror-touch synaesthesia, shared how her heightened empathy makes it difficult to manage others' pain: *"I've also had it more intense where if someone is very hurt, it's very hard for me to deal with because I feel it like I actually feel it."* Similarly, P3, who chose to share her experiences on social media, described her struggles with synaesthesia: *"I'm still struggling for sure. Like a lot of people who watch my videos on Instagram or especially on YouTube [say] 'Ohh, it's a superpower. It's a gift.' I tend to have problems associating synaesthesia as a gift or a superpower because I have a lot of downsides to it, like overstimulation, and I get sick from it. And it's also very confusing, like I'm always confused."* P3 discussed how she developed strategies to cope with these challenges: *"So, like earplugs, if I'm out by myself, noise-canceling headphones. If I am with friends, I'll use the other kind of earplugs so I can still hear them without having to, you know, block everything."*

These challenges seem to also extend into professional contexts. P5, who experiences auditory-olfactory synaesthesia, described how her condition affects her work as a professional taster: *"As a professional taster, I need to be able to taste all the flavors, and for a long time it made my professional life very hard because if the wrong sound was around me, then I would take a sip of wine and it just tastes like pond water with acid in it."* Interestingly, several participants noted that synaesthesia often shaped their career choices, personalities, and interests. P5's professional journey exemplifies how synaesthesia can guide decisions in professional life. Similarly, synaesthesia influenced P9's academic journey: *"I feel like synaesthesia affects all of my passions and study subjects."* Participants also reported using their synaesthesia creatively. For example, P1 described incorporating synaesthetic experiences into her writing: *"I experience it a lot when I write. And I've actually written some characters that have synaesthetic experiences, and I've written sort of metaphors about them into some of the things that I've written, which has been kind of interesting."*

Participants also described how synaesthesia could influence every aspect of their lives, including intimacy, and intersect with other conditions, such as neurodivergence, to shape their experiences. For example, P3, who has ADHD, explained: *"When I do get locked into something, I go all the way. I get hyper-focused on it and just forget everything around me. I just kind of go ahead and go for it. So if I think back to my childhood now, I don't feel like [it was] a negative experience."* P5 reflected on how her relationships are shaped by her synaesthesia: *"I think back on all the relationships I had, this is one thing they all have in common. Because of the voice. Yeah, the voice has to feel good."* These findings underscore the dual nature of synaesthesia, as both a source of unique opportunities and challenges. Synaesthesia influences individuals' personalities, interests,

and personal and professional choices. It also drives them to develop a range of coping mechanisms, allowing them to maximise its benefits and mitigate its drawbacks.

4.3 Importance of self-reflection over time

In navigating the complexities of synaesthesia, highlighted above, our participants emphasised the importance of taking time for self-reflection, learning about their condition, and sharing knowledge with others. This process of understanding and recognising the many dimensions of their synaesthesia often spanned years and required significant personal effort. P12 described the long journey it took to identify the various forms of her synaesthesia, emphasising the gradual development of self-awareness and knowledge: *"The other forms all happen in my mind's eye, so I think my reference point was it has to be that extreme to be something. So it took me another 15 years or so. When I was 25, I looked into it more, and then over the last couple of years, I did more research and talked to clients, podcast guests, and people on Instagram. That's when I realised more and more forms."*

P5 similarly noted how understanding his condition involved deep engagement with various experiences and literature: *"You can learn all of these things through colour theory, right? And after you know color theory, you start seeing the Blues and the Greens and the shadows that, before, they were always there. But you never thought about that because you needed the context. You need the context to notice. And I think that for much of my childhood, you know that when you go into a red room, everything's gonna look different, but you never think about how it looks different."* Discovering synaesthesia can take time because it often feels "normal" to those who experience it. Many participants shared that they initially believed their experiences were universal until others pointed out otherwise. For example, P7 explained: *"I always said it since childhood. Like my personal experience [...] I recently discovered that it's actually a thing. Yeah. That not everybody experienced."* Yet, it seems that only when sufficient self-reflection and time of self-discovering takes place that synaesthetic experiences could be fully harnessed. For examples P1 described how time and experience helped her manage and work with her synaesthesia: *"I think something that I feel like I've come to articulate better as an adult because it's also one of the ways I manage my synaesthesia and have sort of harnessed it for my creative pursuits, is giving myself sensory stimulation helps me focus better."* These findings highlight that discovering one's synaesthesia is rarely a straightforward process. For many participants, the journey of self-discovery is an ongoing process requiring significant commitment and engagement over time. Moreover, taking the time to fully understand and reflect on their synaesthesia is crucial to making the most of it, as emphasised in the previous themes, to enhance their lives and personal well-being.

5 Discussion

In this paper, we explore how direct engagement with synaesthetes, combined with Interpretive Phenomenological Analysis (IPA), can deepen our understanding of synaesthesia and how these insights can be leveraged within HCI. Unlike psychology, which often excludes outliers to focus on synaesthesia as a neurological condition [12, 32, 35], our research centres on the unique and diverse

experiences of synaesthetes, many of whom fall outside conventional diagnostic criteria. HCI research on synaesthesia has typically focused on multisensory interaction and cross-sensory research [15, 18, 20, 25, 31]. However, these approaches often lack the depth necessary to fully capture the complexity of synaesthetic experiences beyond sensory interaction. Our study focuses on the nuanced details of these experiences, revealing that synaesthesia extends beyond immediate sensory perceptions to influence broader aspects of life, including community and environment, personal and professional pursuits, and long-term self-reflection.

A recurring challenge for synaesthetes is isolation, driven by limited public understanding and a lack of accessible information about their condition. Many synaesthetes express a strong desire to foster supportive communities where they can share experiences and access resources. However, the highly individual nature of synaesthesia complicates efforts to address the diverse needs of all members within a single community. Additionally, creating tailored environments that minimise sensory interference is crucial for supporting engagement with synaesthesia but presents significant design challenges. Synaesthesia also profoundly impacts personal and professional life, often requiring individuals to develop unique coping mechanisms and adaptive strategies. This process demands continuous observation, reflection, and effort. Furthermore, understanding and articulating synaesthetic experiences requires time and introspection. The development of articulation-assistive tools, such as a library of terminology for sensory expression, could assist synaesthetes in expressing their perceptions more effectively and foster mutual understanding between synaesthetes and non-synaesthetes.

To address these insights, we are conducting a series of co-design studies to investigate how verbal and non-verbal tools — such as creative and sensory materials, including VR tools and text-to-image/sound AI systems — can enhance the articulation of synaesthetic experiences. These findings aim to inform the design of cross-sensory interaction systems and interactive tools to better support the synaesthetic community. Over time, our research seeks to create a comprehensive library of synaesthetic experiences and toolkits, empowering synaesthetes and those who suspect they may have synaesthesia to better understand, articulate, and navigate their condition.

6 Conclusion

We employed qualitative methods to address two key objectives: 1) to provide an understanding of synaesthesia that embraces its full diversity, and 2) to explore how the insights we report can guide interaction designers in creating more inclusive systems to support people with synaesthesia. Through an in-depth exploration of synaesthetic experiences in the context HCI, we gathered detailed insights via interviews with synaesthetes and analysed the data using Interpretive Phenomenological Analysis. Our analysis identified three key themes that highlight the challenges synaesthetes face in their daily lives - challenges that extend beyond their immediate sensory experiences. Given that synaesthesia remains poorly understood by wider society, fostering a deeper understanding of synaesthetes' experiences and creating spaces where they can comfortably articulate and share their sensory perceptions is

essential. Such efforts not only empower synaesthetes to embrace and articulate their perceptual experiences but also inform the development of interaction designs that incorporate and reflect their unique needs and experiences in HCI.

Acknowledgments

We would like to thank members of the Synaesthesia List, the UK Synaesthesia Association, and the American Synesthesia Association for their generosity in providing resources and contacts. We thank the participants in our studies.

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